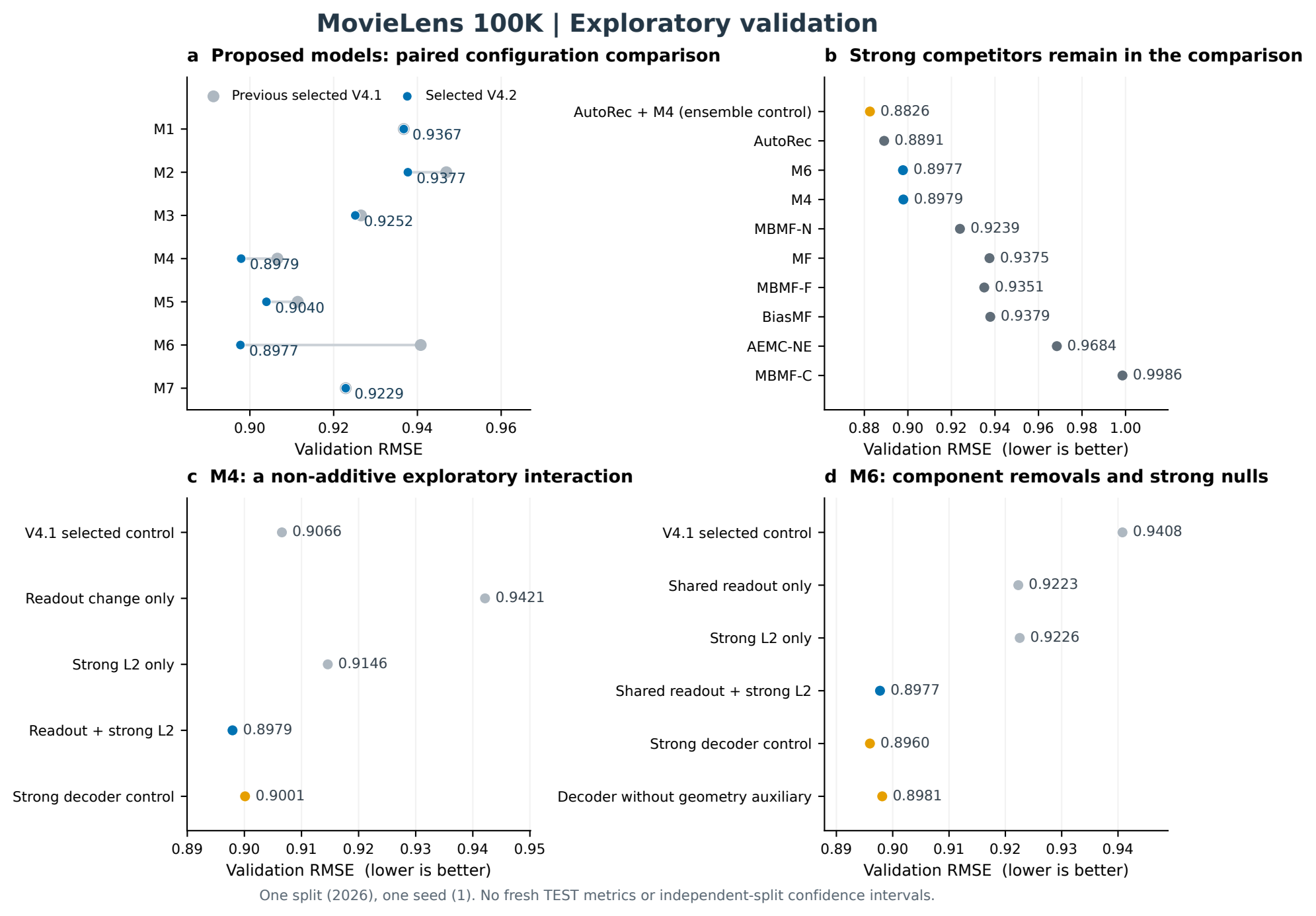
MovieLens 100K | Exploratory validation
a Proposed models: paired configuration comparison
Previous selected V4.1
Selected V4.2
M1
M2
M3
M4
M5
M6
M7
0.9367
0.9377
0.9252
0.8979
0.9040
0.8977
0.9229
0.90
0.92
0.94
0.96
Validation RMSE
b Strong competitors remain in the comparison
AutoRec + M4 (ensemble control)
AutoRec
M6
M4
MBMF-N
MF
MBMF-F
BiasMF
AEMC-NE
MBMF-C
0.8826
0.8891
0.8977
0.8979
0.9239
0.9375
0.9351
0.9379
0.9684
0.9986
0.88
0.90
0.92
0.94
0.96
0.98
1.00
Validation RMSE (lower is better)
c M4: a non-additive exploratory interaction
V4.1 selected control
Readout change only
Strong L2 only
Readout + strong L2
Strong decoder control
0.9066
0.9421
0.9146
0.8979
0.9001
0.89
0.90
0.91
0.92
0.93
0.94
0.95
Validation RMSE (lower is better)
d M6: component removals and strong nulls
V4.1 selected control
Shared readout only
Strong L2 only
Shared readout + strong L2
Strong decoder control
Decoder without geometry auxiliary
0.9408
0.9223
0.9226
0.8977
0.8960
0.8981
0.89
0.90
0.91
0.92
0.93
0.94
Validation RMSE (lower is better)
One split (2026), one seed (1). No fresh TEST metrics or independent-split confidence intervals.